

01 - Hosting Platform, Billing & Infrastructure Strategy

LIONGATE SARL - Internal Technical Handbook

Version 1.0 - Initial release

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Overview

LIONGATE SARL is a startup engineering company that builds internal products and client projects from requirements to production. The infrastructure strategy documented here governs all hosting decisions, billing expectations, environment structures, and domain management practices.

Guiding Principles

Principle	Description
Cost-awareness	Every project must have an estimated monthly cost before provisioning
Environment isolation	Dev, staging, and production are always separate
Right-sizing	Resources are sized for actual workload, not theoretical maximum
Reproducibility	Infrastructure is defined as code and can be recreated from scratch
Security-first	No server is exposed unnecessarily; firewall rules are always restrictive
Backup-mandatory	Every production environment has automated, tested backups

Hosting Provider: Hetzner

Why Hetzner

Hetzner is a German hosting provider offering excellent price-to-performance for European and international deployments. It is LIONGATE SARL's primary infrastructure provider.

Advantage	Detail
Price	Among the lowest EUR-priced VPS/bare metal in Europe
GDPR compliance	German/Finnish data centers, EU data sovereignty
Bare metal	Dedicated servers available at competitive prices
Cloud API	Full REST API and Terraform provider available
Object storage	S3-compatible storage via Hetzner Storage Box or S3 API
Private networks	VLAN-level isolation between servers
Snapshots	Per-server snapshot support for quick recovery

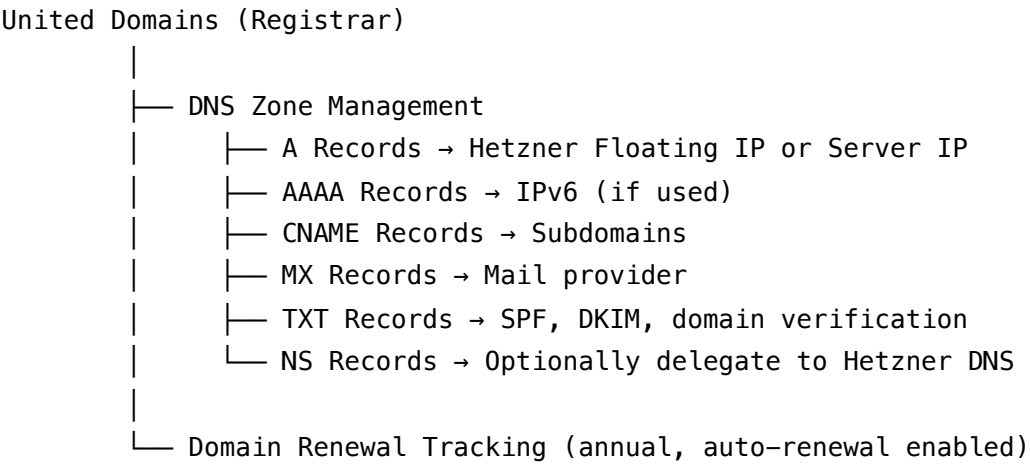
Hetzner Product Lines

Product	Use case	Price range
Cloud VPS (CX/CPX series)	Dev, staging, small production	€3.29–€55/mo
Cloud VPS (CCX series)	CPU-intensive production	€12–€130/mo
Bare Metal (AX/EX/SB)	High-performance production	€35–€220/mo
Managed databases	PostgreSQL, MySQL (optional)	€20–€120/mo
Load balancers	Multi-server HA setups	€5–€40/mo
Storage Box	Backup / SFTP / NFS storage	€3.43–€58/mo
Volumes	Block storage attachable to VPS	€0.052/GB/mo
Snapshots	VM snapshots for recovery	€0.0119/GB/mo
Floating IPs	Static IPs, reattachable	€2/mo each
Object Storage (S3)	File/media/backup storage	€0.012/GB/mo

Domain Provider: United Domains

All LIONGATE SARL domains are registered and managed through **United Domains** (`united-domains.de` / `united-domains.com`).

Domain Workflow



DNS Management Options

Option	Where	When to use
Manage DNS in United Domains	United Domains control panel	Simple projects, single server
Delegate to Hetzner DNS	Point NS records to Hetzner DNS	Complex setups, IaC-managed DNS
Delegate to Cloudflare	Point NS records to Cloudflare	CDN, DDoS protection, proxy needed

Standard DNS Record Types

Record	Example	Purpose
A	app.client.com → 95.216.x.x	Map subdomain to server IP
CNAME	www → app.client.com	Alias record
TXT	v=spf1 ...	Email auth, domain verification
MX	mail.client.com	Email routing
CAA	letsencrypt.org	Restrict SSL certificate authorities

Domain Naming Convention

```
# Production
app.clientname.com
api.clientname.com
admin.clientname.com

# Staging
staging.clientname.com
staging-api.clientname.com

# Internal tools
erp.liongate.com
git.liongate.com
monitor.liongate.com
```

Infrastructure Service Catalog

Compute (Cloud Servers / VPS)

Hetzner Cloud Server Tiers

Tier	vCPU	RAM	Storage	Price/mo	Use case
CX22	2	4 GB	40 GB	€4.35	Dev, lightweight staging
CX32	4	8 GB	80 GB	€8.21	Staging, small production
CX42	8	16 GB	160 GB	€17.77	Medium production
CX52	16	32 GB	320 GB	€35.54	Heavy production
CCX23	4	16 GB	160 GB	€22.75	CPU-dedicated medium
CCX33	8	32 GB	240 GB	€45.18	CPU-dedicated large
CCX43	16	64 GB	360 GB	€90.35	CPU-dedicated XL

Bare Metal (for high-performance use)

Server	CPU	RAM	Storage	Price/mo
AX41-NVMe	AMD Ryzen 5 3600	64 GB	2x NVMe 512 GB	~€37
AX52	AMD Ryzen 9 3900	128 GB	2x NVMe 1.92 TB	~€65
EX44	Intel i9-9900K	128 GB	2x NVMe 1.92 TB	~€70

Note: Bare metal provisioning takes 24–48 hours. Plan ahead.

Database Options

Option	Engine	Price/mo	Notes
Self-hosted on VPS	PostgreSQL 16	€0 extra	Included in server cost
Hetzner Managed DB	PostgreSQL	€20–€120	HA option, managed patches
Separate DB server	PostgreSQL 16	€8–€35	Dedicated DB VPS

LIONGATE default: Self-hosted PostgreSQL on a dedicated DB server per production environment.
Hetzner Managed DB is an option for clients requiring SLA-backed managed services.

Storage & Volumes

Service	Type	Price	Use
Hetzner Volumes	Block storage	€0.052/GB/mo	Persistent Docker volumes, DB data
Storage Box (BX11)	100 GB NAS/SFTP	€3.43/mo	Dev/staging backups
Storage Box (BX31)	1 TB NAS/SFTP	€8.69/mo	Production backups
Storage Box (BX61)	5 TB NAS/SFTP	€34.45/mo	Large backup archives
Object Storage	S3-compatible	€0.012/GB/mo	Static assets, media, archives

Load Balancers

Type	Price/mo	Max targets	Notes
LB11	€5.39	5	Small setups
LB21	€10.79	25	Mid-scale
LB31	€21.59	50	Production

Networking

Resource	Price	Notes
Inbound traffic	Free	No charge
Outbound traffic (within Hetzner)	Free	Between servers on private network
Outbound traffic (internet)	Included in server plan (20 TB+ typical)	
Private network (VLAN)	Free	Internal communication
Floating IP	€2/mo	Static, reassignable IP
IPv6	Free	Per server

SSL / TLS

Method	Cost	Use case
Let's Encrypt (Certbot)	Free	Default for all servers
Wildcard cert (Let's Encrypt DNS challenge)	Free	*.client.com multi-subdomain
Commercial SSL	€50–€300/yr	Enterprise client requirement only

Environment Tiers

Every project at LIONGATE SARL follows a structured environment model. Each environment has its own infrastructure and is **never shared with another project**.

Environment Overview

Environment	Purpose	Data	Access	Cost level
Development (dev)	Local dev / feature work	Synthetic / seeded	Internal devs	Minimal (often local)
Staging	Integration testing, UAT	Anonymized prod copy	Internal + client review	Low
Production	Live user-facing system	Real data	Controlled	Medium–High
High-performance production	High-traffic or SLA-bound	Real data	Controlled	High
Low-cost production	Internal tools, low traffic	Real data	Controlled	Low

Environment Infrastructure Profiles

Development Environment

Local machine or shared dev server
Docker Compose (all services locally)
No external access required
Seeded / synthetic data only

Recommended: Docker Desktop / local compose

External hosting: Optional – CX22 if shared team dev server needed

Cost: €0 (local) or ~€4/mo (shared dev VPS)

Staging Environment

Hetzner Cloud VPS (CX32 or CX42)
App + DB on same server or separate
SSL enabled, domain: staging.project.com
Accessible to internal team + client
Reset/seed data regularly

Cost: €8–€20/mo per project

Standard Production Environment

App Server: Hetzner CX42 (8 vCPU, 16 GB RAM)
DB Server: Hetzner CX32 (4 vCPU, 8 GB RAM)
Volume: 100 GB block storage for DB data
Backup: Storage Box BX31 (1 TB)
Floating IP: Yes
SSL: Let's Encrypt
Monitoring: Self-hosted (Grafana stack)

Cost: ~€30–€55/mo

High-Performance Production Environment

	App Server: Hetzner CCX33 or bare metal AX41	
	DB Server: CCX23 (dedicated CPU, 16 GB RAM)	
	Load Balancer: LB21	
	Volume: 200–500 GB block storage	
	Backup: Storage Box BX61 (5 TB)	
	Floating IP: Yes (2 for HA)	
	Monitoring: Grafana + alerting	
	Optional: Redis cache server (CX22)	

Cost: ~€100–€200/mo

Low-Cost Production Environment

	Single server: Hetzner CX32 (app + DB)	
	Volume: 50 GB block storage	
	Backup: Storage Box BX11 (100 GB)	
	SSL: Let's Encrypt	
	Monitoring: Lightweight (UptimeRobot free tier)	

Cost: ~€12–€18/mo

Best for: Internal tools, admin dashboards, low-traffic sites

Project Type Profiles

Each project type has defined infrastructure needs. This section maps project type to infrastructure.

Profile 1 - Standard Web Application

Backend: Node.js / Python / etc.

Frontend: React / Vue (served as static or via server)

Database: PostgreSQL

Traffic: Low to medium (< 10k req/day)

Recommended setup:

- App + reverse proxy: CX32 or CX42
- Database: Separate CX22 or included on same server
- Storage: 40–80 GB volume
- Backup: Storage Box BX11
- SSL: Let's Encrypt
- Monitoring: Basic

Estimated monthly cost: €15–€35

Profile 2 - Database-Heavy Application

Backend: API-heavy, complex queries, large datasets

Database: PostgreSQL with large schema and significant storage

Recommended setup:

- App server: CX42
- DB server: CX42 or CCX23 (dedicated CPU)
- Volume: 200–500 GB
- Backup: Storage Box BX31 + S3
- Read replica: Optional CX32

Estimated monthly cost: €45–€100

Profile 3 - Odoo Instance

Odoo ERP / CRM deployment for internal use or client

Recommended setup:

- App server: CX42 or CCX23 (Odoo is CPU/RAM-intensive)
- DB server: CX32 or CX42 dedicated
- Volume: 100–300 GB (for filestore + DB)
- Backup: Storage Box BX31
- Workers: 4–8 Odoo workers
- Cron: 2 dedicated workers

Minimum viable: CX42 all-in-one (8 vCPU, 16 GB RAM)

Recommended: App CX42 + DB CX32 separated

Estimated monthly cost: €30–€70

See: 02-odoo-deployment-docker-image.md and 03-odoo-deployment-github-source.md

Profile 4 - Internal Enterprise Tools

Admin panels, dashboards, internal portals

Recommended setup:

- CX32 all-in-one
- Small volume (20–50 GB)
- Basic backup

Estimated monthly cost: €10–€18

Profile 5 - High-Traffic Client Project

E-commerce, SaaS product, platform with real-time features

Recommended setup:

- 2+ App servers behind load balancer
- Dedicated DB server (CCX series)
- Redis cache
- CDN (Cloudflare free tier or paid)
- Volume: 200+ GB
- Backup: BX61 + S3
- Monitoring: Full Grafana/Prometheus stack
- CI/CD: GitHub Actions + auto-deploy

Estimated monthly cost: €120–€250+

Pricing Model & Cost Estimates

Monthly Cost Summary Table

Environment	Compute	DB	Storage	Backup	LB	SSL	Monitoring	Total/mo
Dev (local)	€0	€0	€0	€0	€0	€0	€0	€0
Dev (shared VPS)	€4	incl.	incl.	€0	€0	Free	€0	~€4
Staging	€8–€18	incl.	€2–€5	€3.43	€0	Free	€0	~€13–€26
Prod (low)	€8	incl.	€5	€3.43	€0	Free	Free tier	~€17
Prod (standard)	€18–€36	€8–€18	€5–€10	€8.69	€0	Free	€5	~€40–€70
Prod (high-perf)	€45–€90	€23–€45	€10–€26	€34	€11	Free	€10	~€130–€215

Hidden Costs to Account For

Cost	Frequency	Notes
Domain registration	Annual	€10–€30/year per domain
Floating IP	Monthly	€2/IP - assign per production environment
Snapshot storage	Monthly	€0.0119/GB × snapshot size
Bandwidth overage	Rare	>20 TB outbound (unlikely at startup scale)
Hetzner Managed DB premium	Monthly	+€20–€60 vs self-hosted
Cloudflare paid plan	Monthly	€20+/mo if advanced WAF needed
Commercial SSL	Annual	Only if client mandates
External email provider	Monthly	Mailgun / Postmark / Brevo

Annual Budget Estimation Template

Project: [Name]
Type: [Standard / Odoo / High-traffic / Internal]
Environments: [Dev, Staging, Production]

Monthly cost breakdown:

Production: € ____
Staging: € ____
Dev server: € ____
Domains: € ____ / 12 (amortized)
Email: € ____
Other: € ____

Total/month: € ____
Total/year: € ____

Reviewed by: [Name]
Date: [Date]

Infrastructure Selection Rules

Use these decision rules when provisioning new infrastructure.

Server Selection Decision Tree

Is this a development environment?

- └─ YES → Docker locally or shared CX22 dev server
- └─ NO → Continue

Is this a staging environment?

- └─ YES → CX32 (combined) or CX22+CX22 (split)
- └─ NO → Continue

Is this production?

- └─ YES → Apply production rules below

What is the expected traffic / resource profile?

- └─ Low (< 1k users/day, internal tools)
 - └─ CX32 all-in-one + BX11 backup
- └─ Medium (1k–50k users/day, standard app)
 - └─ CX42 app + CX32 db + BX31 backup
- └─ High (50k+ users/day, SaaS/e-commerce)
 - └─ CCX33 app × 2 + LB + CCX23 db + BX61 backup
- └─ Odoo ERP
 - └─ CX42 app + CX32 db + 100 GB volume + BX31 backup

Database Placement Rules

Scenario	Decision
Dev / staging	DB on same server as app (simpler)
Production (any)	DB on dedicated server
Production with SLA	Consider Hetzner Managed DB
High write load	Dedicated CCX DB server
Read-heavy	Add read replica

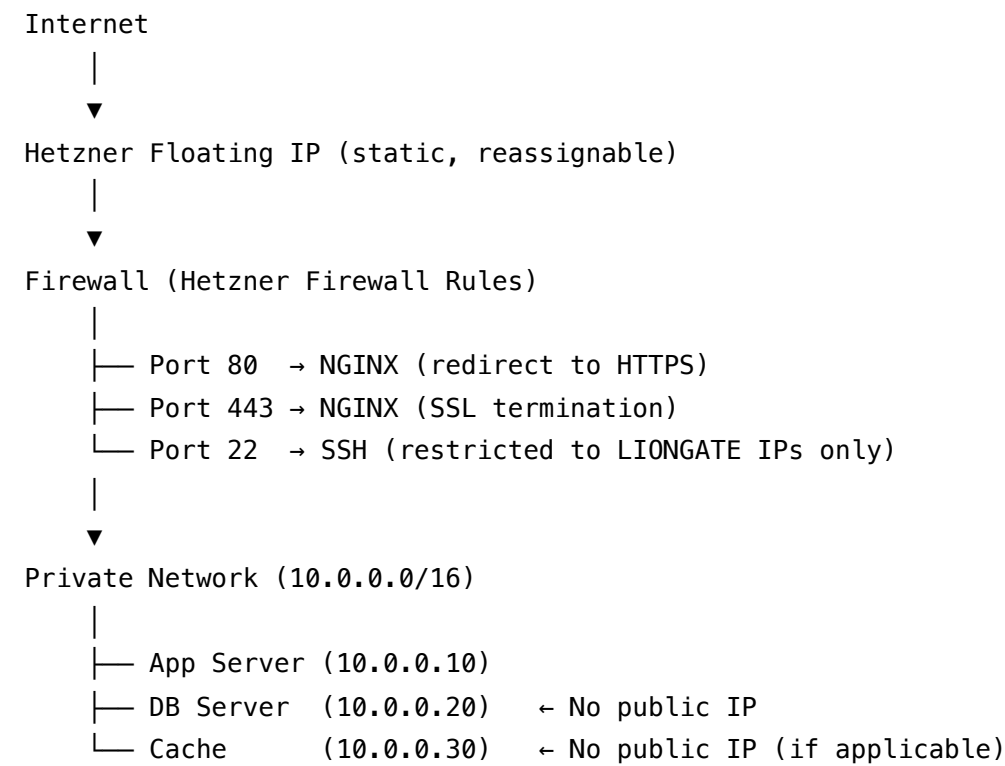
Volume Rules

Data type	Volume?	Size
PostgreSQL data directory	Mandatory in production	2× expected DB size
Odoo filestore	Mandatory	50 GB minimum

Data type	Volume?	Size
Application logs	Optional	20–50 GB
Media / uploads	Mandatory	Based on project

Network & Security Architecture

Network Architecture Per Environment



Hetzner Firewall Standard Rules

```
# Inbound allowed
- TCP 22    → Source: LIONGATE office IPs only (whitelist)
- TCP 80    → Source: 0.0.0.0/0 (redirect to 443)
- TCP 443   → Source: 0.0.0.0/0

# Inbound blocked (everything else)
- TCP 5432  → BLOCKED from internet (PostgreSQL)
- TCP 6379  → BLOCKED from internet (Redis)
- TCP 8069  → BLOCKED from internet (Odoo direct)

# Outbound
- Allow all (servers need to reach package repos, etc.)
```

Backup & Disaster Recovery Strategy

Backup Tiers

Tier	Frequency	Retention	Target	Method
Database snapshots	Daily	7 days	Storage Box	pg_dump via cron
Database snapshots	Weekly	4 weeks	Storage Box	pg_dump via cron
Database snapshots	Monthly	12 months	S3/Object Storage	pg_dump compressed
VM snapshots	Weekly	2 snapshots	Hetzner Snapshots	Hetzner API
Filestore / uploads	Daily	7 days	Storage Box	rsync
Application config	On change	Git history	GitHub private repo	Manual + CI

Backup Storage Sizing Guide

DB size × 7 daily = weekly storage needed (uncompressed)
Apply 60% compression ratio for pg_dump .gz
Add 20% overhead

Example: 20 GB DB
Daily compressed dump: ~8 GB
7-day retention: ~56 GB
Monthly archive: ~8 GB × 12 = ~96 GB/yr
→ Storage Box BX31 (1 TB) sufficient for most projects

Recovery Time Objectives

Scenario	RTO	RPO	Method
Server crash	15–30 min	24h	Restore VM snapshot, restore from backup
DB corruption	30–60 min	24h	Restore pg_dump to clean DB
Accidental data deletion	1–4h	24h	Restore from daily dump
Complete datacenter loss	2–8h	24h	Rebuild from IaC + restore DB backup

Backup Testing Schedule

- Monthly: Restore one DB backup to a test server and validate
- Quarterly: Full disaster recovery drill (rebuild from scratch)
- Document results in internal runbook

Monitoring & Observability

Monitoring Stack

Tool	Purpose	Deployment
Prometheus	Metrics collection	Docker on monitoring server
Grafana	Dashboards	Docker on monitoring server
Node Exporter	Server metrics	On every server

Tool	Purpose	Deployment
Postgres Exporter	DB metrics	On DB servers
Nginx Exporter	HTTP metrics	On app servers
Loki	Log aggregation	Docker on monitoring server
Alertmanager	Alert routing	Docker on monitoring server
UptimeRobot	External uptime check	Free tier (5 monitors)

Standard Alerts

Alert	Threshold	Severity
CPU usage	> 85% for 5 min	Warning
CPU usage	> 95% for 2 min	Critical
Memory usage	> 80%	Warning
Disk usage	> 75%	Warning
Disk usage	> 90%	Critical
DB connection pool	> 80%	Warning
HTTP 5xx rate	> 1%	Warning
HTTP 5xx rate	> 5%	Critical
Backup job failure	Any failure	Critical
SSL cert expiry	< 14 days	Warning

Scaling Strategy

Vertical Scaling (Scale Up)

Resize the server to a larger plan in Hetzner. This requires a brief downtime (5–10 min).

Trigger: CPU/RAM consistently > 80% for 1 week

Action:

1. Snapshot the server
2. Stop the server
3. Resize in Hetzner panel (or API)
4. Start and validate

Downtime: 5–15 minutes

Horizontal Scaling (Scale Out)

Add more app servers behind a load balancer.

Trigger: High traffic, app servers saturated

Requirements:

- Stateless app (no local sessions/files)
- Shared DB server
- Shared file storage (volumes or S3)
- Load balancer configured

Steps:

1. Deploy second app server
2. Configure Hetzner Load Balancer
3. Update health checks
4. Update floating IP to point to LB

Database Scaling

Read-heavy: Add read replica PostgreSQL

Write-heavy: Scale DB server vertically (CCX series)

Data volume: Add/resize Hetzner volume

Connection pooling: Use PgBouncer in front of PostgreSQL

Infrastructure-as-Code Standards

All LIONGATE infrastructure must be defined as code. This enables:

- Reproducibility
- Audit trail
- Onboarding efficiency
- Disaster recovery

Tooling

Tool	Purpose
Terraform	Provision Hetzner servers, networks, firewalls, volumes
Ansible	Configure servers (OS setup, packages, services)
Docker Compose	Application stack definition
GitHub Actions	CI/CD pipelines
Vault / .env + Secret Manager	Secrets management

Repository Structure

```
infrastructure/
├─ terraform/
│   ├─ environments/
│   │   ├─ staging/
│   │   └─ production/
│   └─ modules/
│       ├─ server/
│       ├─ database/
│       ├─ network/
│       └─ backup/
│   └─ README.md
├─ ansible/
│   ├─ playbooks/
│   │   ├─ base-server.yml
│   │   ├─ install-docker.yml
│   │   └─ configure-nginx.yml
│   └─ inventory/
└─ README.md
```

Cost Control Policies

Rules

- 1. No server is provisioned without a cost estimate in the project ticket
- 2. Dev servers are destroyed or stopped when not in use

3. **Staging servers are stopped outside business hours** (saves ~60% on compute)
4. **VM snapshots are limited** - old ones must be deleted after each rotation
5. **Monthly billing review** is mandatory - alert if actual > estimate by > 20%
6. **No floating IP is left unattached** - they still cost €2/mo

Monthly Review Checklist

- ☐ Review Hetzner billing dashboard
- ☐ Check for orphaned servers, volumes, snapshots
- ☐ Validate that stopped servers are stopped (staging)
- ☐ Compare actual spend to project budget
- ☐ Archive or delete backups beyond retention policy
- ☐ Renew domains expiring within 60 days
- ☐ Review SSL certificate expiration dates

Document owner: LIONGATE SARL IT/DevOps Team

Last updated: See Git history

Next review: Quarterly